3/8/26

EXPERIMENT - 4

Setting up a local area Network

AIM:

To design, set up and configure a local area network connecting 4 computers through a network switch, assign static IP address to all devices verify connectivity using the ping command.

IP configuration table:

Device	IP address	Subnet network	Connected to
PC0	125.1.1.0	255.0.0.0	Switch Fo/1
PC1	125.1.1.1	255.0.0.0	Switch Fo/2
PC2	125.1.1.2	255.0.0.0	Switch Fo/3
PC3	125.1.1.3	255.0.0.0	Switch Fo/4
Switch	125.1.1.4	255.0.0.0	VLAN1 Interface.

Implementation:

Step 1: Open cisco packet tracer and place switch, 4 PCs and one server in the workspace.

Step 2: Connect all PCs and the server to the switch using copper straight through cables.

Step 3: Configure all IP address of all PCs and the server so that they belong to same network.

Step 4: Verify that all cable connection are active

Step 5: Test network connectivity by opening the command prompt on a PC using ping command.

Step 6: Enable the HTTP services on the server by selecting server → services → HTTP = ON.

Step 7: Open web browser on PC and enters the server IP address to access the servers default web page.

Step 8: Switch from realtime mode to simulation mode in packet tracker.

Step 9: Verify that successful communication takes place.

RESULT: A LAN was successfully established using switch and a server and packet flow was observed successfull.